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ORIGINAL PAPER - PRODUCTION ENGINEERING

A comprehensive comparative analysis of particle swarm optimization and genetic algorithms in well acidizing optimization

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Abstract

Well productivity plays a vital role in determining the economic viability of hydrocarbon field development. Among stimulation techniques, matrix acidizing is highly effective for improving well performance by removing near-wellbore formation damage. The success of an acidizing treatment depends on the precise selection of injection volume and rate to maximize damage removal while minimizing operational costs. This study introduces an integrated computational framework for optimizing matrix acidizing processes. The framework first predicts post-treatment outcomes and then determines optimal injection parameters. Its main objective is to identify the acid volume and injection rate that minimize the post-treatment skin factor, thereby enhancing well productivity while reducing acid consumption and cost. The methodology consists of two sequential phases: prediction and optimization. In the prediction phase, four machine learning algorithms—Extra-Trees (ExTree), Random Forest, Gradient Boosting, and AdaBoost—were trained using a comprehensive dataset derived from extensive reservoir simulations to estimate post-acidizing skin factor and injection pressure. Among these models, the ExTree algorithm achieved the highest predictive accuracy, with an R^2 value of 0.9390 for the skin factor. In the optimization phase, the validated ExTree model was coupled with two metaheuristic algorithms, Genetic Algorithm (GA) and Particle Swarm Optimization (PSO), to determine the optimal injection parameters. Both optimization algorithms consistently converged to stable solutions within 500 iterations. The framework's computational cost is practical for engineering design and offline analysis, confirming its feasibility for field applications. The novelty of this research lies in its unified, data-driven approach that combines high-accuracy prediction with advanced optimization techniques. This integration provides a robust and efficient tool for the rational design of well stimulation treatments.

Keywords Matrix acidizing · Optimization · Genetic algorithm (GA) · Extra-Trees · Particle swarm optimization (PSO)

Abbreviations

AB	AdaBoost
ANN	Artificial Neural Networks
CI	Cohort Intelligence
DT	Decision Tree
DTS	Distributed Temperature Sensing
ET	Extra Trees
FBP	Formation Breakdown Pressure
FFNN	Feedforward Neural Network(s)
GA	Genetic Algorithm
GBDT	Gradient Boosted Decision Trees
GB	Gradient Boosting
GENFIS2	Generalized Fuzzy Inference System
KNN	K-Nearest Neighbors
LREG	Linear Regression
MAE	Mean Absolute Error
MAM	Matrix Acidizing Model
ML	Machine Learning
MLP	Multilayer Perceptron
MSE	Mean Squared Error
MMSCFD	Million Standard Cubic Feet per Day
NNs	Neural Networks
PSO	Particle Swarm Optimization
PVBT	Pore-Volume-to-Breakthrough
RF	Random Forest
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination (or R-squared)
SVM	Support Vector Machine(s)

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SVR Support Vector Regression

Introduction

Carbonate reservoirs constitute a major source of global hydrocarbon production, hosting roughly 60% of the world's oil and 40% of its gas reserves (Hatamizadeh and Sedaee 2023; Li et al. 2025a, b). Despite their economic importance, these formations frequently suffer from permeability impairment caused by drilling and cementing operations. Such near-wellbore damage can significantly reduce well productivity and hinder efficient fluid flow. Acidizing is one of the most common and effective stimulation techniques used to restore permeability in damaged formations. It involves the injection of acidic solutions to dissolve pore-blocking materials near the wellbore, thereby improving fluid mobility and overall formation performance (Hatamizadeh and Sedaee 2023; Li et al. 2018, 2024; Abbasi et al. 2024). Among acidizing approaches, matrix acidizing is particularly notable because it is conducted at pressures below the fracture threshold. This allows for selective dissolution of damage without inducing new fractures in the reservoir rock (Araújo et al. 2024; Keihani Kamal et al. 2024). A key objective of matrix acidizing is to reduce the skin factor, a dimensionless quantity that represents the additional pressure drop caused by near-wellbore damage (Ali et al. 2016; Cruz-Maya et al. 2011). To achieve an optimal treatment, engineers must determine appropriate values for acid volume and injection rate. These parameters greatly influence the dissolution efficiency and post-treatment productivity. Traditionally, their selection has relied on labor-intensive laboratory testing or computationally expensive numerical simulations (Keihani Kamal et al. 2024; Dai et al. 2024). However, both conventional approaches are often limited by their high cost, slow turnaround time, and restricted applicability to heterogeneous reservoir conditions.

Recent progress in machine learning (ML) has introduced a transformative approach to optimizing acidizing processes in well stimulation (Yousefmarzi et al. 2025). By utilizing historical well and reservoir data, ML models can uncover complex relationships among various parameters—such as lithology, fluid properties, and operational conditions—that are difficult to capture through traditional analytical or simulation-based methods (Yousefmarzi et al. 2024; Pankaj et al. 2018; Bigdeli and Delshad 2023, 2024; Campos et al. 2024; Li et al. 2025). The use of optimization frameworks, including those developed for reservoir scheduling, demonstrates the scalability and practical potential of ML-driven methodologies (Liu et al. 2025). These models can accurately predict optimal acid compositions, injection rates, and pressures customized for specific well conditions. Beyond improving operational efficiency, such data-driven strategies significantly reduce trial-and-error costs and enhance the reliability of acidizing treatments (Kellogg et al. 2018). The integration of ML into acidizing workflows therefore represents a vital step toward data-centric well stimulation, offering wide applicability across diverse carbonate reservoirs.

Vazquez et al. (2016) investigated the influence of overflush fluids on the longevity of scale squeeze treatments. Using stochastic optimization algorithms, particularly Particle Swarm Optimization (PSO), they optimized key design parameters including inhibitor concentration, main slug volume, overflush volume, and shut-in time (Vazquez et al. 2016). Recent advances in Distributed Temperature Sensing (DTS) technology have significantly improved the optimization of matrix acidizing treatments in carbonate formations. Livescu et al. (2018) developed a novel model that integrates DTS data to enhance acid placement through optimized treatment scheduling and wormhole modeling. By combining DTS data with comprehensive matrix acidizing simulations, researchers achieved better predictions of acid injection rates and volumes, leading to improved well productivity and reduced operational costs (Livescu et al. 2018). Sidaoui et al. (2018) employed machine learning (ML) techniques to predict the optimal injection rate for carbonate acidizing operations. Using a dataset comprising 170 experimental observations, they applied a Support Vector Machine (SVM) model to capture the relationship between various input parameters and the dimensionless pore-volume-to-breakthrough (PVBT). The SVM model demonstrated the highest predictive accuracy, achieving an R^2 value of 0.8773, which corresponded to the lowest prediction error among the tested algorithms. It outperformed other ML approaches, including the Generalized Fuzzy Inference System and Feedforward Neural Networks (FFNN), confirming its superior capability in modeling nonlinear relationships and predicting key acidizing performance metrics (Sidaoui et al. 2018). Dong et al. (2022) focused on fracturing parameter optimization in tight oil reservoirs using machine learning and evolutionary algorithms. They developed and compared four models—Support Vector Regression (SVR), Gradient Boosted Decision Trees (GBDT), Random Forest (RF), and Multilayer Perceptron (MLP)—to predict production performance. Among these, the MLP model combined with PSO achieved the highest net present value (NPV), demonstrating superior capability for optimizing hydraulic fracturing operations. Although GBDT and RF also performed well, the MLP's high accuracy underscores the value of ML integration in production optimization (Dong et al. 2022). Muther et al. (2022) extended this line of research to tight gas reservoirs, employing Neural Networks (NN) combined with socio-inspired optimization algorithms, including Cohort Intelligence (CI), Multi-Cohort Intelligence, Teaching–Learning-Based Optimization (TLBO), PSO, and Genetic Algorithms (GA). Their integrated framework coupled reservoir simulation with these advanced algorithms to maximize gas production efficiency. The model achieved outstanding predictive performance, with R^2 values of 0.999 for training and 0.998 for testing (Muther et al. 2022). Hatamizadeh and Sedaee (2023) pioneered the use of machine learning (ML) for modeling and optimizing acidizing performance in carbonate reservoirs. They evaluated a range of ML algorithms, including Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Linear Regression (LREG), Multilayer Perceptron (MLP), Decision Tree (DT), Artificial Neural Networks (ANN), and meta-learning approaches such as stacking. Their results demonstrated the versatility and effectiveness of data-driven methods for stimulation design. The developed core-scale model achieved an impressive 83% prediction accuracy for acid pore volume to breakthrough (PVBT). Furthermore, their optimization approach, based on Genetic Algorithm (GA) techniques, delivered substantial improvements in acidizing efficiency, confirming the potential of ML–GA integration for reservoir stimulation optimization (Hatamizadeh and Sedaee 2023). Dargi et al. (2023) extended ML–GA integration for acidizing design optimization across multiple Iranian reservoirs. They analyzed 218 acidizing operations using models based on genetic programming. Their genetic programming approach achieved an R^2 of 0.82 and an RMSE of 17.65 surpassing the predictive performance of Artificial Neural Networks, Random Forest, and XGBoost models (Dargi et al. 2023). Deng et al. (2025) emphasized the critical importance of fractures, rock matrix, and reservoir permeability in determining gas reservoir productivity. Similarly, numerical simulation of residual oil saturation underscores the importance of post-stimulation fluid dynamics (Deng et al. 2025).

This study addresses the challenge of predicting and optimizing the post-acidizing skin factor, a key parameter influencing well productivity enhancement. The primary objective is to develop a comprehensive predictive–optimization framework that integrates advanced machine learning (ML) algorithms with metaheuristic optimization techniques. This framework aims to accurately predict the post-acidizing skin factor and determine optimal acidizing parameters—including injected acid volume and rate—to minimize formation damage and maximize treatment efficiency. Four state-of-the-art ML algorithms—Random Forest, Extra Trees, Gradient Boosting, and AdaBoost—are combined with Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) to tackle the complex, nonlinear nature of acidizing processes. Unlike previous studies that largely depended on conventional statistical approaches or limited field data, the present work employs a simulation-based data generation strategy, producing a robust and diverse dataset for model training and validation. This approach reveals intricate relationships between critical input parameters and post-acidizing outcomes, achieving unprecedented prediction accuracy and reliability. Although acidizing has been extensively investigated, few studies have systematically integrated predictive modeling with optimization for real-time decision-making in field operations. The proposed framework therefore provides actionable guidance for selecting acidizing parameters under varying reservoir conditions, contributing to both theoretical understanding and practical advancement in well stimulation optimization.

Methodology

This section outlines the systematic methodology employed to achieve the research objectives, emphasizing the prediction and optimization of acidizing parameters for oil and gas reservoirs. The proposed approach integrates data preparation, machine learning (ML) techniques, and optimization algorithms to accurately predict key outputs, including the post-acidizing skin factor and acid injection pressure.

Through a combination of simulation-based analysis, advanced ML models, and metaheuristic optimizers, the study seeks to enhance both the efficiency and operational safety of acidizing treatments.

Data selection and generation

The foundation of an effective predictive and optimization framework is the selection of appropriate input parameters. An initial survey of literature and industry practices identified a comprehensive set of variables known to influence matrix acidizing outcomes. These included reservoir properties (e.g., pressure, temperature, permeability, porosity), formation characteristics (e.g., mineralogy, heterogeneity), completion details, and treatment specifications (e.g., acid type, volume, rate, additives).

From this extensive list, a focused subset of 16 parameters was selected for model development to ensure the resulting framework is both robust and practical for field application. The selection process was governed by the following criteria:

- **Primary influence**: Parameters demonstrated to have a first-order impact on the post-acidizing skin factor were prioritized. This was guided by fundamental petroleum engineering principles and preliminary sensitivity analyses, which confirmed that primary injection parameters (volume, rate) and fundamental reservoir characteristics (e.g., pre-acidizing permeability, damage diameter) were the dominant controlling factors across a wide range of scenarios.

- **Data availability and reliability**: The selection was constrained to parameters for which data can be reliably and consistently obtained in typical field operations, enhancing the model's practical utility.
- **Model parsimony**: The number of variables was carefully managed to create a parsimonious model that avoids overfitting, reduces computational complexity, and improves interpretability.

The final model incorporated the following parameters: formation type, reservoir type, well completion method, reservoir pressure, pre-acidizing permeability, porosity, fracture gradient, reservoir temperature, pre-acidizing skin, type of acid solution, damage diameter, formation breakdown pressure, volume and rate of injected acid, acid injection pressure, and post-acidizing skin.

Several parameters, while important, were held constant or excluded from this specific analysis. These included acid additives, detailed formation mineralogy, and fine-scale heterogeneity. For example, acid additives (such as corrosion inhibitors, iron control agents, and surfactants) are critical for the operational success and integrity of a treatment, but their effect is often tailored to specific fluid-rock interactions and downhole conditions. Their impact, while significant, is typically considered a secondary optimization step after the primary treatment parameters (volume and rate) have been established. Including them would have required a multi-fold increase in the dimensionality of the problem, making the systematic analysis of the core parameters intractable. Therefore, this study deliberately focused on establishing a robust model for the primary drivers of damage removal. This approach isolates the most influential factors and clarifies their interactive effects on the final skin factor. A detailed quantitative analysis of the impact of specific additive packages represents a logical next step and is recommended for future research.

The study utilized a comprehensive dataset generated through numerical simulations to train machine learning models. Realistic ranges of values were defined for each selected parameter to reflect actual reservoir and well conditions. Using these ranges, 441 experimental designs were created, representing diverse acidizing scenarios. An advanced commercial acidizing simulator was employed to run these scenarios and calculate the corresponding post-acidizing skin factors, ensuring the physical and chemical interactions were modeled with high fidelity.

Machine learning models and optimization algorithms

To predict the post-acidizing skin factor, four ensemble learning models were employed: Random Forest, Extra Trees, Gradient Boosting, and AdaBoost. These models were selected over simpler approaches due to their ability to effectively capture the complex, non-linear interactions inherent in reservoir stimulation processes while maintaining a lower risk of overfitting. The bagging-based models (Random Forest and Extra Trees) reduce prediction variance by averaging multiple decision trees, thus improving model stability. In contrast, the boosting-based models (Gradient Boosting and AdaBoost) build decision trees sequentially, correcting errors from previous iterations, resulting in higher predictive accuracy. To optimize the acidizing parameters, two metaheuristic algorithms—Genetic Algorithm (GA) and Particle Swarm Optimization (PSO)—were incorporated. These optimization techniques efficiently explore the complex search space of acidizing variables to identify optimal operational conditions. Model performance was rigorously evaluated using standard statistical indicators, including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2), ensuring a comprehensive assessment of predictive reliability (Chicco et al. 2021).

Random forest (RF)

Random Forest is an ensemble learning method based on the principle of bootstrap aggregating, or "bagging," which constructs a multitude of decision trees during training. The algorithm's robustness stems from a dual randomization strategy. First, each tree is trained on a unique bootstrap sample of the data (i.e., a random sample selected with replacement). Second, and more critically for model generalization, at each node of a tree, only a random subset of input features is considered for determining the best split. This second step, feature randomness, directly addresses the issue of overfitting. By preventing any single, highly predictive feature from systematically dominating the structure of all trees, the algorithm forces the ensemble to explore a more diverse range of predictive relationships. This process effectively "decorrelates" the individual trees, meaning their individual errors are less likely to be systematic. Consequently, when the predictions from all trees are averaged to produce the final output—such as the estimated skin factor—the individual model errors tend to cancel each other out. This averaging process significantly reduces the overall model variance and mitigates the risk of overfitting to noise in the training data, a common challenge with single decision trees (Kolarik et al. 2024; Smith et al. 2013; Zhang et al.

2018; Pedregosa et al. 2011). Furthermore, the algorithm's structure allows for the quantification of feature importance by measuring the mean decrease in impurity when a given feature is randomly permuted, providing valuable insight into which operational parameters most influence the stimulation outcome (Breiman 2001; Dehghani et al. 2024).

Extra trees

The Extra Trees algorithm is a variant of Random Forest that introduces an additional layer of randomization to further reduce model variance. Like RF, it builds an ensemble of decision trees from the training data. However, it differs in two key aspects: (1) it uses the entire original training sample to grow each tree, forgoing the bootstrap sampling step, and (2) it selects the split point at each node in a fully randomized manner. Instead of calculating the optimal split point for a feature, it selects a random split point. This extreme randomization often results in slightly higher bias but can effectively reduce variance, making it computationally efficient and robust, particularly for datasets with many features where some may be noisy or irrelevant (Ghazwani and Begum 2023; Geurts et al. 2006).

Gradient boosting

Gradient Boosting is a sequential ensemble technique that builds the model in a stage-wise fashion. It synthesizes a final strong model from a series of weak learners, typically decision trees. The process begins by fitting an initial simple model to the data. Subsequently, a new weak learner is trained at each iteration to predict the pseudo-residuals (the gradient of the loss function) of the preceding model. The core principle is functional gradient descent; each new tree is added to the ensemble to take a step in the direction that best minimizes the overall loss function (e.g., Mean Squared Error (MSE) for regression). The final prediction is the sum of the predictions from all learners in the sequence, often weighted by a learning rate parameter () to prevent overfitting. The iterative nature of Gradient Boosting allows it to correct the errors of its predecessors, leading to highly accurate and powerful predictive models (Bentéjac et al. 2021; Larestani et al. 2022; Zhang et al. 2018; Shehadeh et al. 2021; Dorogush et al. 2018; Huang et al. 2019; Tao et al. 2024; Kizayev et al. 2023).

Table 1 Hyperparameter settings for the Genetic algorithm (GA)

Hyperparameter	Value
Population Size	50
Number of generations	500
Mutation rate	0.1
Mutation Lype	Uniform mutation
Crossover Late	0.8
Crossover Lype	Two-point Crossover
Selection Lethod	Tournament Selection
Elitism Lount	2

AdaBoost

AdaBoost is another sequential ensemble method that adaptively adjusts the weighting of training instances at each iteration. Unlike Gradient Boosting, which fits subsequent models to residuals, AdaBoost focuses on prediction errors. It begins by assigning equal weights to all data points in the training set. In each iteration, a weak learner is trained, and the weights of the data points that were incorrectly predicted by that learner are increased. Consequently, the subsequent weak learner is forced to focus more on these "difficult" examples. The final prediction is a weighted sum of the predictions from all weak learners, where the weights are determined by each learner's training accuracy. While powerful, AdaBoost's mechanism of up-weighting difficult samples makes it more sensitive to noise and outliers in the dataset compared to methods like Random Forest (Patil et al. 2018).

Genetic algorithm (GA)

This stochastic optimization algorithm is highly efficient and suitable for problems with multiple objectives. GA operates on a population of potential solutions (chromosomes) and evolves them through selection, mutation, and crossover operations. It requires minimal parameter settings and can be combined with other algorithms for improved optimization performance (Rahmanifard and Plaksina 2019). For the optimization of acid treatment parameters in this work, the GA was implemented using the specific hyperparameters listed in Table 1.

Particle swarm optimization (PSO)

Inspired by the collective behavior of bird flocks or fish schools, PSO is another optimization algorithm. It starts with a randomly generated population of potential solutions (particles) and iteratively updates their positions based on their own best performance and the best performance observed by their neighbors. Unlike GA, PSO does not utilize mutation and crossover operators (Rahmanifard and Plaksina 2019; Souza et al. 2022).

Prediction of acid injection pressure and Post-Acidizing skin factor

Once trained and validated, the machine learning models were employed to predict acid injection pressure for different injection rates and post-acidizing skin factors. A crucial consideration in this step was ensuring that the predicted injection pressure remained within the maximum allowable limit to prevent formation damage. Maintaining this constraint is essential for preserving reservoir integrity and ensuring safe and efficient operations.

Optimization of acidizing parameters

To enhance the acidizing process, key operational parameters—namely, injected acid volume (bbl) and injection rate (bbl/min)—were optimized. These parameters were treated as decision variables within an optimization framework, with the objective of minimizing the post-acidizing skin factor. Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) were employed due to their efficiency in exploring complex, non-linear solution spaces and identifying near-optimal solutions.

The optimization process adhered to the following constraints to ensure practical feasibility:

- Post-acidizing skin factor: 0 to − 4.
- Injected acid volume: 4000 to 25,000 gallons.
- Injected acid rate: 1 to 10 bbl/min.
- Maximum acid injection pressure: 300 psi below the formation fracture pressure (Yu et al. 2024).

Model comparison and validation

The optimization results obtained from PSO and GA were compared to identify the most effective approach for minimizing the post-acidizing skin factor. This comparison provided insights into the robustness and reliability of each optimization method. To confirm the practical applicability of the findings, the machine learning predictions and optimization results were validated using commercial simulation software.

Results and discussion

The dataset comprised 441 acidized wells, which were divided into training, validation, and testing subsets to ensure robust model development and evaluation. A total of 352 wells (80%) were assigned to the training set, among which 70 wells (20% of the training set) were used for validation. The remaining 89 wells (20%) constituted the testing set, serving to assess the model's generalization capability. Two distinct modeling approaches were employed to predict the post-acidizing skin factor and acid injection pressure. Following the prediction phase, two optimization algorithms were applied to determine the optimal injected acid volume and injection rate, with the goal of minimizing the post-acidizing skin factor. Further details of the modeling and optimization procedures are provided in the subsequent sections.

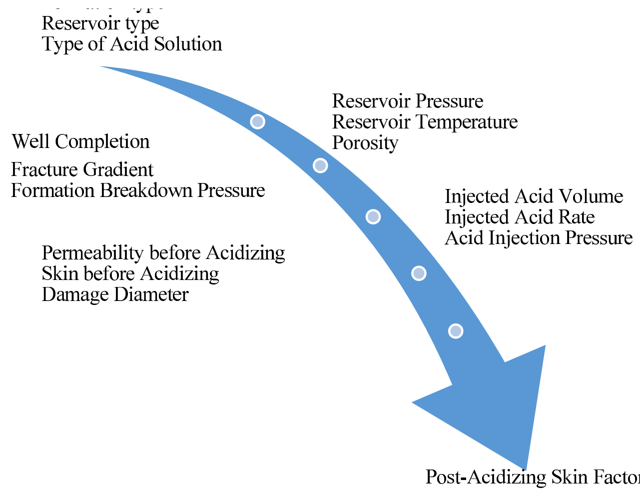

Fig. 1 Input and output parameters of the post-acidizing skin factor model

Table 2 Error metrics and accuracy of each model for Post-Acidizing skin factor prediction

Algorithms	MAE	MSE	RMSE	R^2
AdBoost	0.281	0.138	0.372	0.9242
Extra-trees	0.196	0.111	0.334	0.9390
RF	0.221	0.116	0.341	0.9364
GB	0.250	0.177	0.421	0.9030

Post- acidizing skin factor prediction model

In this section, the performance of four machine learning models introduced in the methodology section for predicting the post-acidizing skin factor is presented and compared. The aim of these models is to analyze the relationships between parameters and quantify their impacts. The input and output parameters of the models are summarized in Fig. 1.

For this purpose, four algorithms (Random Forest, Extra Trees, Gradient Boosting, and AdaBoost) were employed. The algorithm with the highest accuracy and best performance was selected as the predictive model for the post-acidizing skin factor. The values of MAE, MSE, RMSE, and R^2 were calculated for each algorithm, as shown in Table 2.

Based on comprehensive evaluation, the Extra Trees algorithm was identified as the most accurate model due to its superior performance across critical evaluation metrics, including accuracy, MSE, and R^2 score. It demonstrated the lowest MAE, MSE, and RMSE, along with the highest coefficient of determination (R^2) among the models assessed. These results highlight the algorithm's effectiveness in capturing the complex, non-linear relationships between input

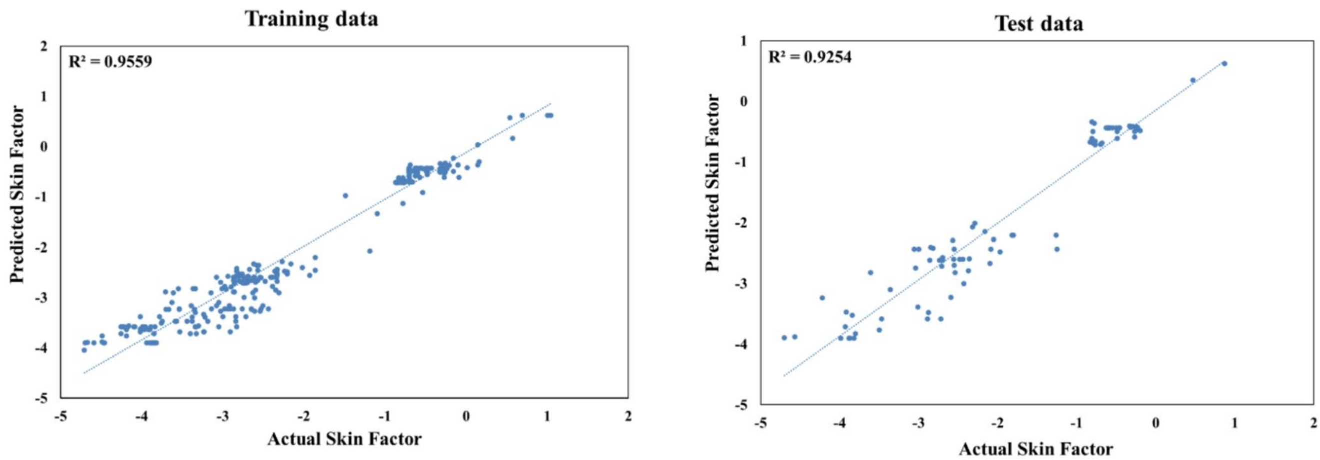

Fig. 2 Scatter plot of AdaBoost predicted vs. Actual skin factor

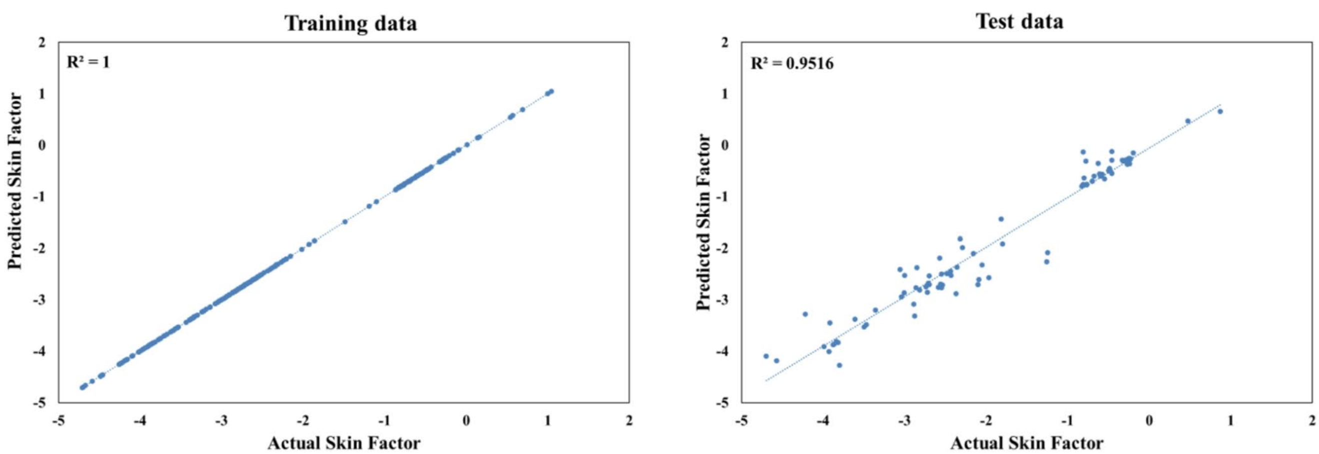

Fig. 3 Scatter plot of extra-trees predicted vs. Actual skin factor

parameters and the post-acidizing skin factor. This selection is further supported by two critical factors. First, acidizing operations involve numerous interacting variables, leading to significant data variability and uncertainty. The inherent randomness of the Extra Trees algorithm enables it to effectively model these complexities. Second, field data often contain noise and outliers due to measurement errors and natural variability. The algorithm's randomization helps mitigate the impact of these inconsistencies, ensuring more reliable predictions. The strong performance of the Extra-Trees model in this context aligns with recent advancements in the application of surrogate models for predicting reservoir performance (Yanchun et al. 2024). Therefore, the Extra Trees algorithm is suggested as a robust and reliable tool for predicting the post-acidizing skin factor.

The distribution and range of the data are visualized in Figs. 2, 3, 4 and 5 through scatter plots comparing actual and predicted values for both the training and testing datasets. These figures illustrate the regions of optimal performance for the methods used. The vertical axis represents the predicted values, while the horizontal axis corresponds to the actual values. Blue markers represent the skin factor values for both the test and training datasets. The scatter plots reveal that the Extra Trees model achieves the closest alignment between the measured and predicted skin factor values. This is demonstrated by the data points clustering near the 45-degree line, which represents a perfect 1:1 correspondence between actual and predicted values.

Many machine learning methods are often considered "black boxes" due to the opaque nature of the relationship between input parameters and outputs. To address this, there is growing interest in explainable machine learning techniques. One such approach is parameter importance analysis, which identifies the most influential input parameters by evaluating the reduction in model accuracy when a particular parameter is omitted. This analysis helps pinpoint inputs that have significant positive or negative impacts on the output (Mohammadian et al. 2022).

The study also presents a box plot (Fig. 6) displaying the distribution of the skin factor data. This graphical

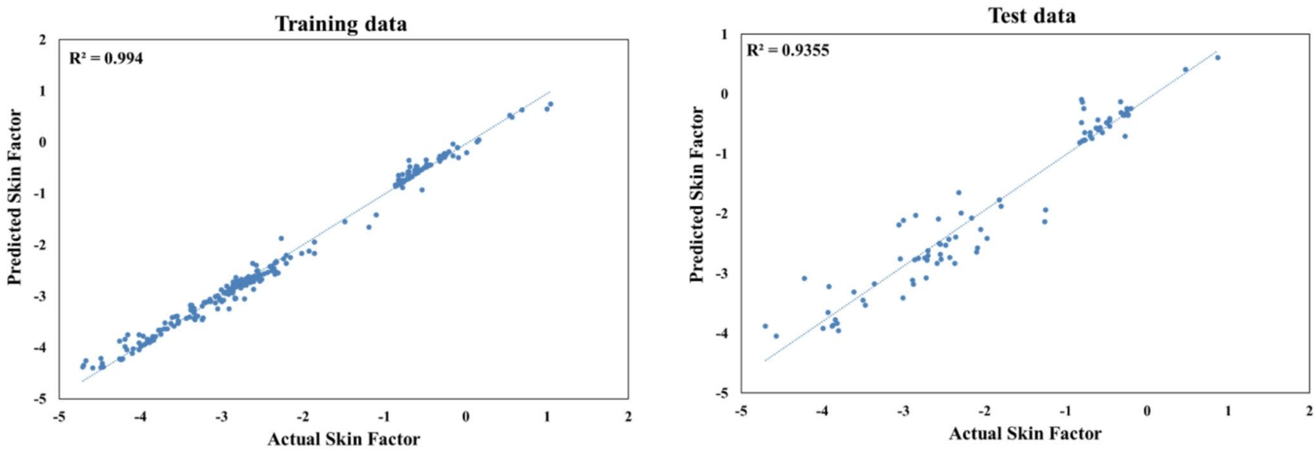

Fig. 4 Scatter plot of random forest predicted vs. Actual skin factor

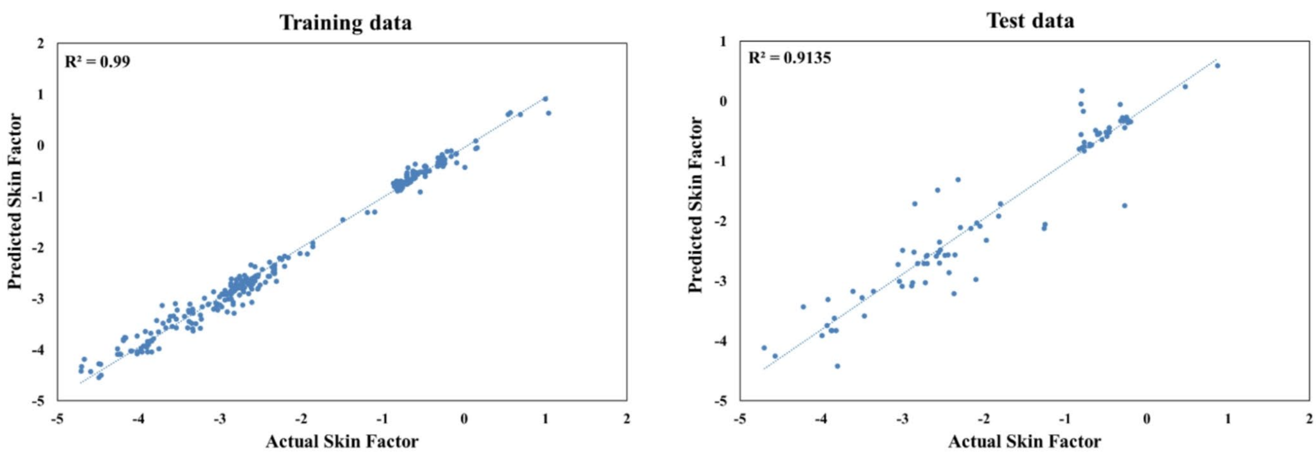

Fig. 5 Scatter plot of gradient boosting predicted vs. Actual skin factor

representation allows for the comparison of the minimum, maximum, and range of values predicted by the different algorithms against the actual skin factor values.

The box plot analysis suggests that the Extra-Trees model has the best coverage of the data range, with its predicted values encompassing a wider spread that more closely matches the true distribution of the skin factor. This indicates the Extra-Trees model's ability to capture the full breadth of the skin factor values, rather than potentially under- or over-predicting certain ranges.

In conclusion, the Extra-Trees model, with its superior performance, interpretability, and ability to capture a broader range of skin factor variations, emerges as the most promising approach for predicting skin factor in this context.

Acid injection pressure prediction model

In this section, before predicting acid injection pressure using machine learning algorithms, the effect of acid injection pressure on acid injection rate was first investigated.

Several factors can influence the pressure-rate relationship, including formation permeability, formation damage, fluid properties, wellbore conditions, and acid properties. Lower permeability formations, such as tight sandstones, require higher pressures to overcome resistance to fluid flow. The presence of formation damage can significantly increase flow resistance and necessitate higher injection pressures. Fluid viscosity and density can influence pressure drop, especially at higher flow rates. Gas reservoirs may exhibit different pressure-flow rate relationships compared to oil reservoirs due to compressibility effects .

Therefore, to optimize acidizing operations, understanding the pressure-rate relationship for each reservoir type is crucial. Excessive injection rates can lead to premature formation breakdown, while insufficient rates may not achieve adequate acid penetration.

Figure 7 illustrates the relationship between the required acid injection pressure and the injection rate for four distinct reservoir scenarios: carbonate oil, carbonate gas, sandstone

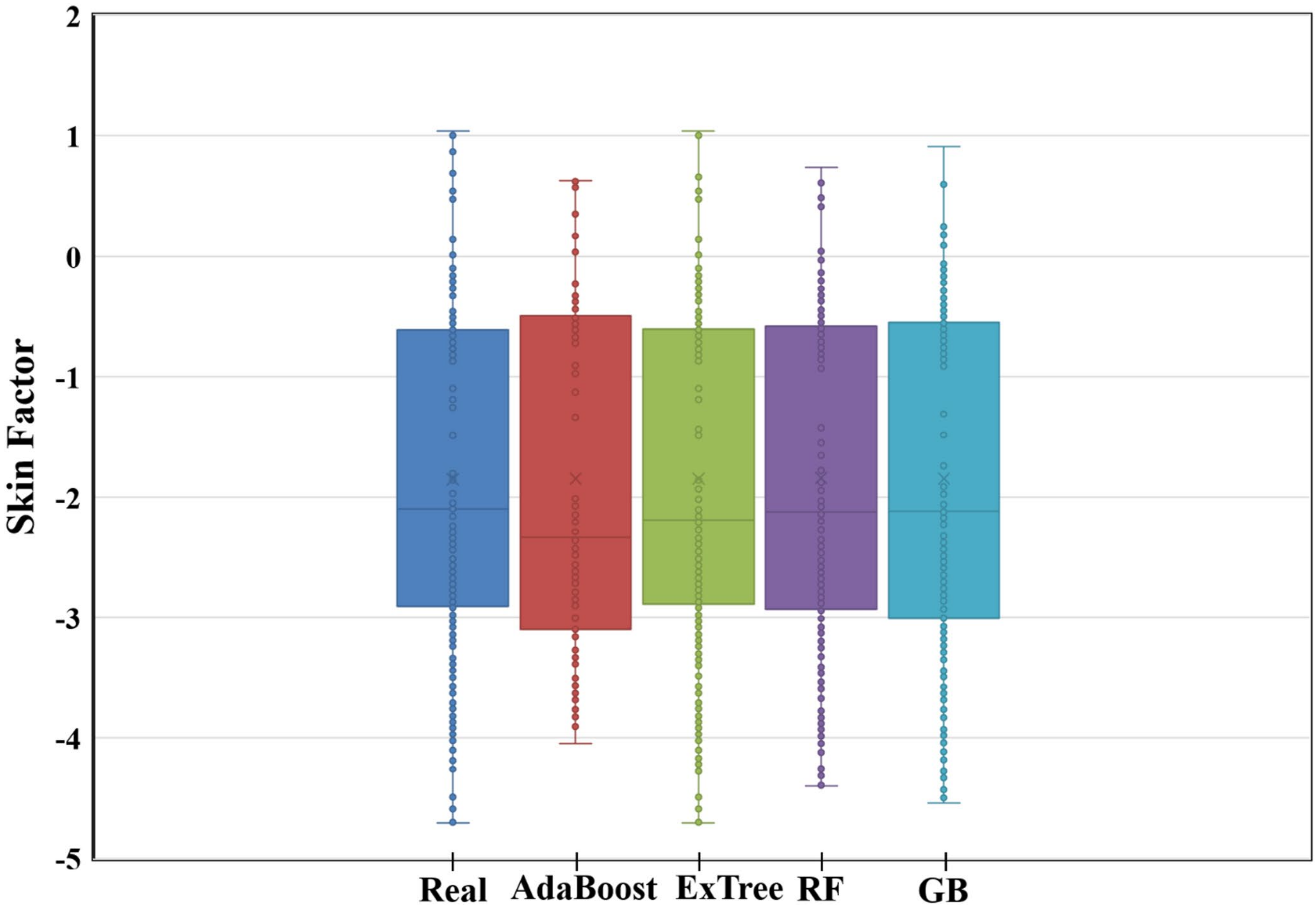

Fig. 6 Box plot of skin factor distribution for different models

oil, and sandstone gas, each under its specific formation breakdown pressure (FBP).

The data in Fig. 7 reveals a complex, non-linear relationship. While there is a general, slight upward trend in injection pressure with increasing injection rate, this trend is superimposed with significant pressure fluctuations. Rather than a simple linear or smooth curve, the injection pressure exhibits an oscillatory or undulating behavior across the range of injection rates for all reservoir types.

This oscillatory pattern is likely the manifestation of a complex interplay of competing physical phenomena occurring simultaneously downhole:

- **Wormhole propagation and branching**: During acidizing, particularly in carbonates, dominant flow paths called 'wormholes' are created. The propagation of these wormholes is not uniform. As a wormhole extends, it can temporarily lower the resistance to flow, causing a pressure drop. However, fluid diversion into secondary branches or encounters with lower permeability zones can increase resistance, leading to a subsequent pressure rise. This continuous process of wormhole creation, branching, and competition for fluid can create pressure oscillations.
- **Formation heterogeneity**: Natural variations in permeability, porosity, and mineralogy along the wellbore mean that as the acid front advances, it continuously encounters different rock properties (Cao et al. 2024). Injecting into a high-permeability streak would require less pressure, while encountering a tighter section would cause a pressure spike, contributing to the observed fluctuations.
- **Fines migration and plugging**: In sandstone formations, the reaction of acid (especially HF acid) with clays and feldspars can release fine particles (fines). These fines can be transported by the injected fluid and clog pore throats, causing a rapid increase in injection pressure. Subsequent pressure buildup may then dislodge these blockages, leading to a sudden pressure drop. This cyclical plugging and un-plugging mechanism is a well-known source of pressure fluctuations during sandstone acidizing.
- **Fluid-rock reaction dynamics**: The rate of chemical reactions between the acid and the rock minerals is

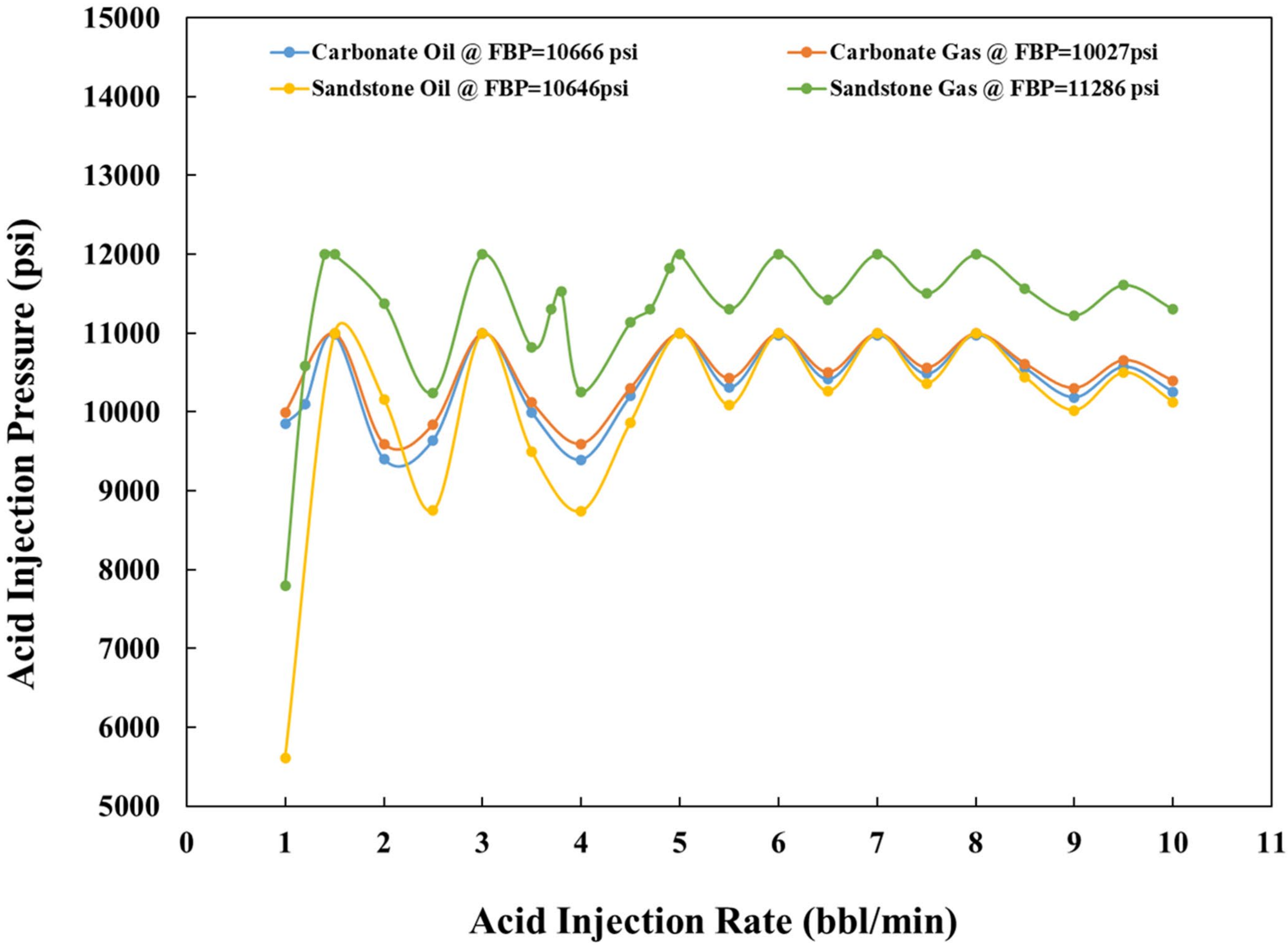

Fig. 7 Acid injection pressure vs. Rate for reservoir types at Formation Breakdown Pressure (FBP)

dependent on factors like temperature and acid concentration, which themselves change along the flow path. These changing reaction rates can alter the near-wellbore permeability in a non-uniform manner, influencing the injection pressure.

The Fig. 7 also reveals that carbonate oil and gas reservoirs, as well as sandstone oil reservoirs, exhibit relatively similar trends. However, sandstone gas reservoirs require significantly higher injection pressures compared to the other three reservoir types. This observation suggests that carbonate formations may have higher permeability or experience less formation damage, enabling easier fluid flow.

The increased formation damage in sandstone reservoirs is likely due to the reaction of the mud acid (hydrochloric and hydrofluoric acids) with certain calcite components of the rock. Specifically, sandstone reservoirs—particularly sandstone gas reservoirs—demand considerably higher injection pressures at comparable flow rates. This indicates greater formation resistance, potentially caused by lower permeability or more severe formation damage.

Following the analysis of the relationship between acid injection pressure and injection rate, this section evaluates the performances of four machine learning models—Random Forest, Extra Trees, Gradient Boosting, and AdaBoost—in predicting acid injection pressure (as detailed in Sect. "Post- acidizing skin factor prediction model"). Figure 8 outlines the input and output parameters considered for these models.

To assess the model accuracies, various performance metrics were calculated, including MAE, MSE, RMSE, and R^2. The results, summarized in Table 3, indicate that the Extra Trees algorithm outperforms the other models. It exhibits the lowest values for MAE, MSE, and RMSE, as well as the highest R^2 value.

To gain insights into the model's predictions and assess its performance visually, scatter plots were generated for both training and testing datasets (Figs. 9, 10, 11 and 12). These plots compare the actual and predicted acid injection pressures. The Extra Trees model consistently demonstrates the closest alignment between predicted and actual values,

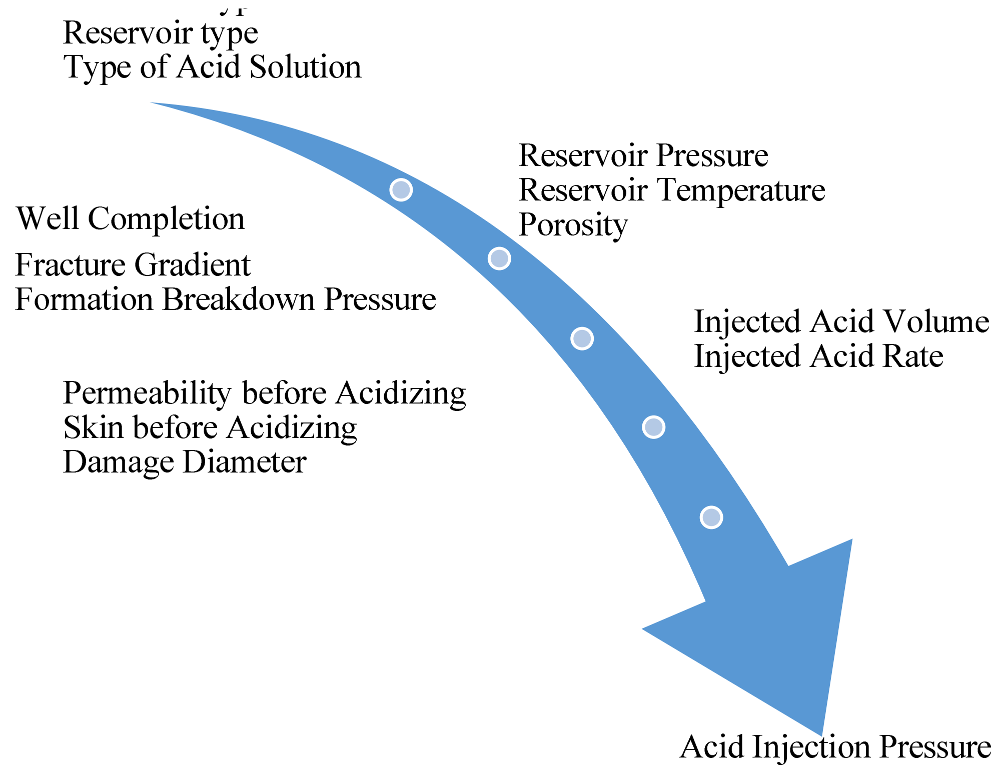

Fig. 8 Input and output parameters of the acid injection pressure model

Table 3 Error metrics and accuracy of each model for acid injection pressure prediction

Algorithms	MAE	MSE	RMSE	R^2
AdBoost	395.22	284604.469	533.483	0.6846
Extra-trees	202.162	151973.894	401.970	0.8207
RF	255.255	168818.290	410.875	0.8129
GB	372.942	246350.379	496.337	0.7288

suggesting its superior ability to capture the underlying patterns in the data.

To further visualize the model performance distribution, a box plot (Fig. 13) is presented. By comparing the minimum and maximum values of the algorithms to the real values, it is evident that Extra-trees covers a wider range of data, indicating its robustness in handling diverse data scenarios.

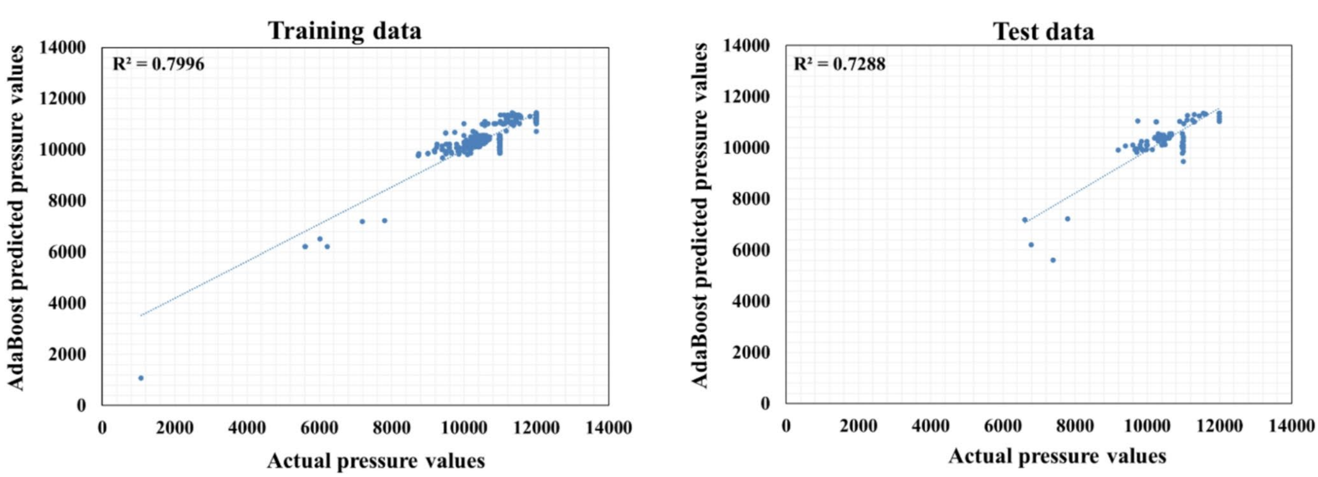

Fig. 9 Scatter plot of AdaBoost predicted vs. Actual injected acid pressure

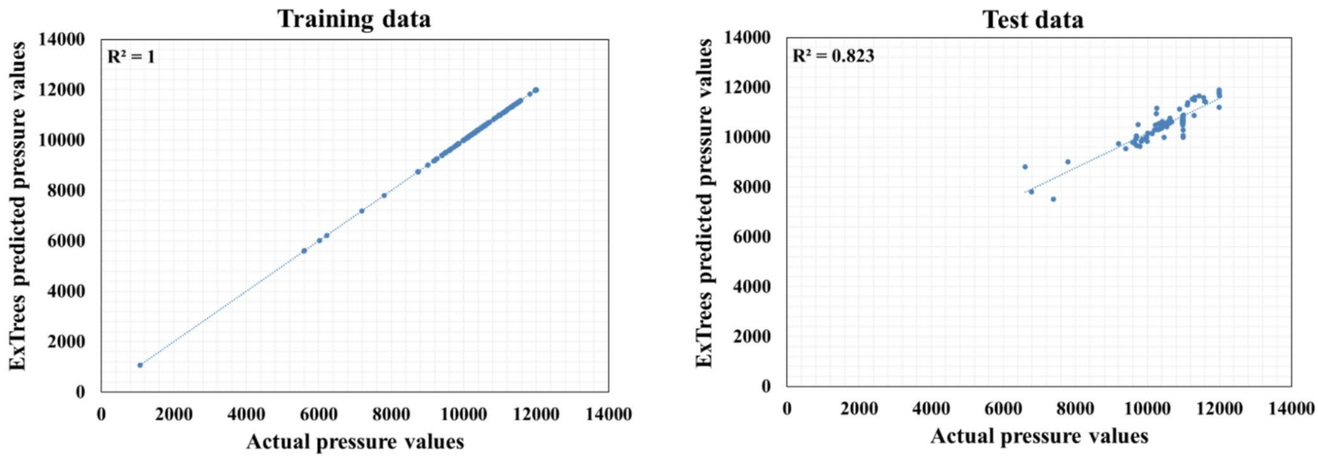

Fig. 10 Scatter plot of extra-trees predicted vs. Actual injected acid pressure

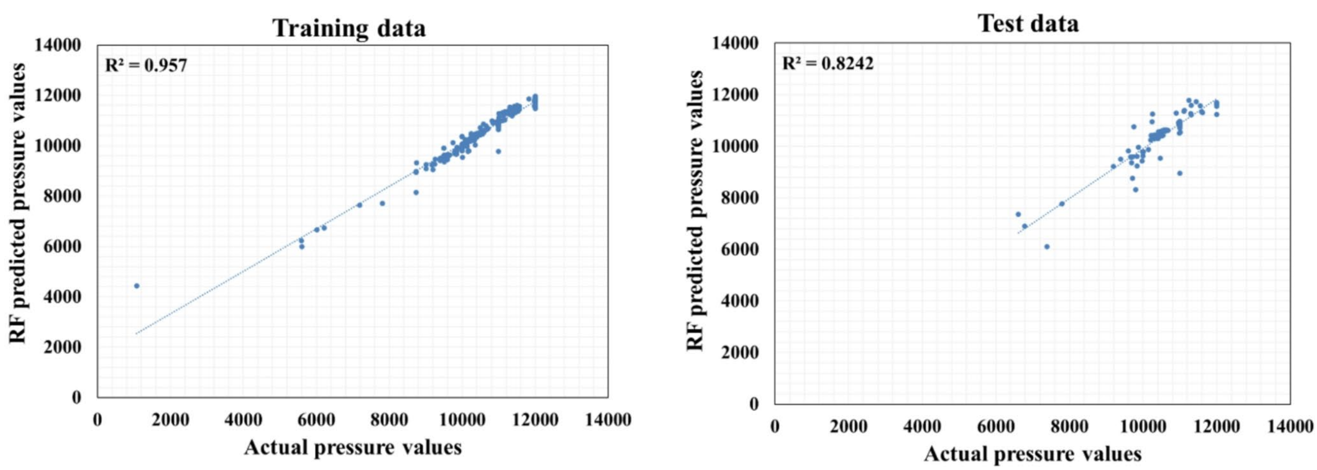

Fig. 11 Scatter plot of random forest predicted vs. Actual injected acid pressure

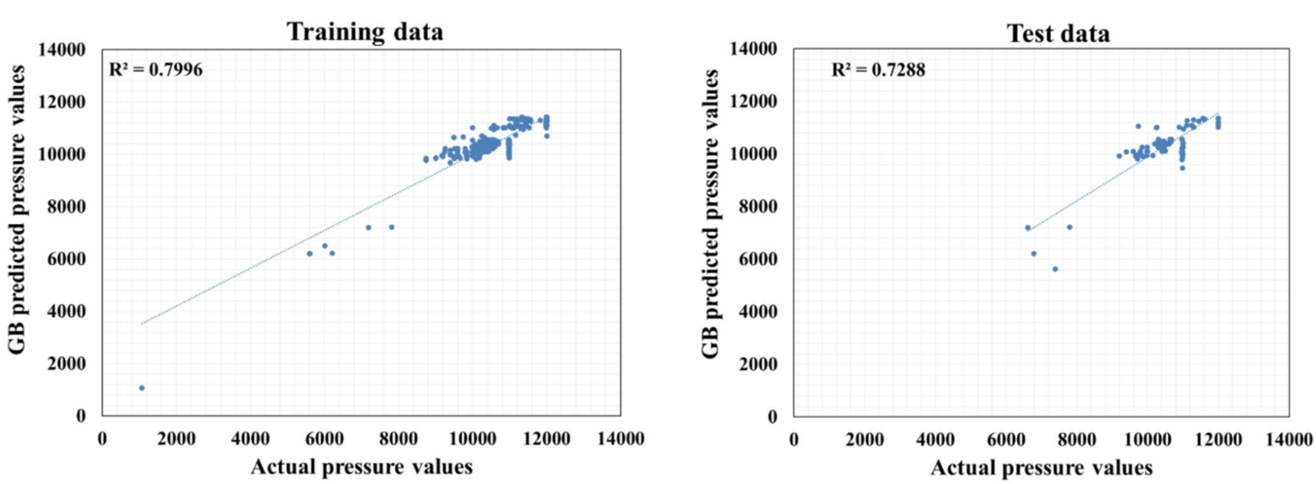

Fig. 12 Scatter plot of gradient boosting predicted vs. Actual injected acid pressure

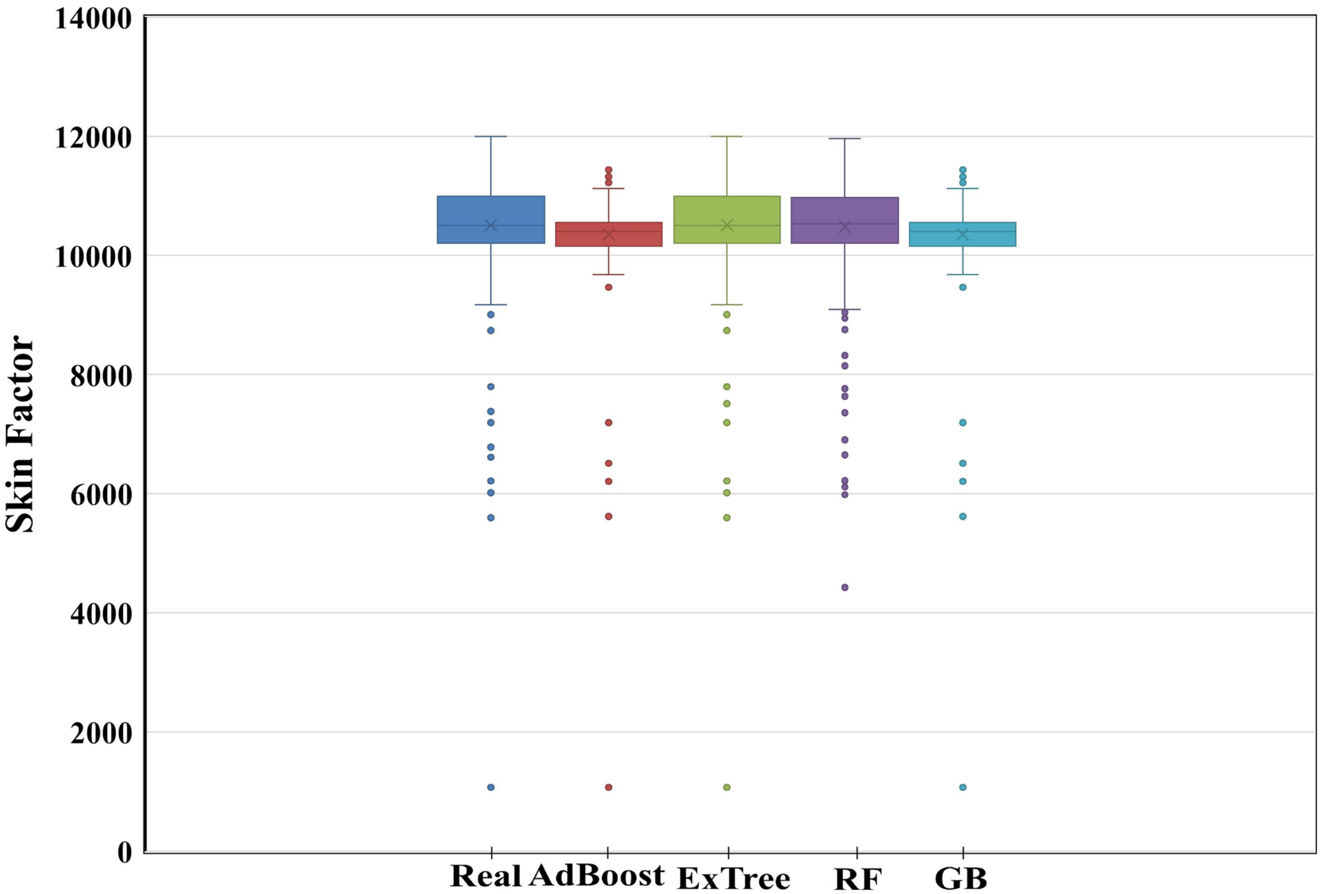

Fig. 13 Box plot of injected acid pressure distribution for different models

Optimization methods

A Genetic Algorithm (GA) was employed to optimize the injected acid volume and rate, aiming to minimize the post-acidizing skin factor. The Extra Trees algorithm was utilized to estimate the post-acidizing skin factor. The optimization problem was constrained by the feasible range of the post-acidizing skin factor (0 to -4). Additionally, the non-linear relationship between acid injection pressure and injection rate was modeled using the Extra Trees algorithm. To assess the performance of the GA, a convergence analysis was conducted. Figure 14 illustrates the convergence process of the objective function (post-acidizing skin factor). The algorithm exhibits rapid convergence in the initial iterations, followed by a gradual stabilization. This indicates the efficiency of the GA in finding optimal solutions. The rapid initial convergence suggests that the GA effectively explores the solution space, while the subsequent gradual stabilization demonstrates its ability to exploit promising regions. While a formal error analysis is beyond the scope of this study, the convergence and stability of the GA are critical performance indicators. The rapid convergence observed in Fig. 14 suggests that the algorithm is efficient in finding optimal solutions. Additionally, the consistent behavior of the GA across multiple iterations indicates its stability.

A Particle Swarm Optimization (PSO) algorithm was employed to optimize the injected acid volume and rate, aiming to minimize the post-acidizing skin factor. Similar to the GA, the Extra Trees algorithm was utilized to predict the post-acidizing skin factor and acid injection pressure.

Figure 15 presents the convergence process of the PSO for a representative well. The PSO also demonstrates rapid initial convergence, followed by a stabilization phase. This behavior is indicative of the PSO's efficiency in exploring the solution space and converging to optimal solutions.

Both Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) have demonstrated effectiveness in minimizing the post-acidizing skin factor. The GA exhibits rapid initial convergence, followed by a gradual stabilization, indicating efficient exploration and exploitation of the solution space. PSO also shows rapid initial convergence, followed by a relatively quick stabilization, suggesting efficient exploration and exploitation. While GA is more robust to noise and local optima and can handle complex, non-linear problems, it can be computationally expensive and may have slower convergence. PSO, on the other hand, is faster

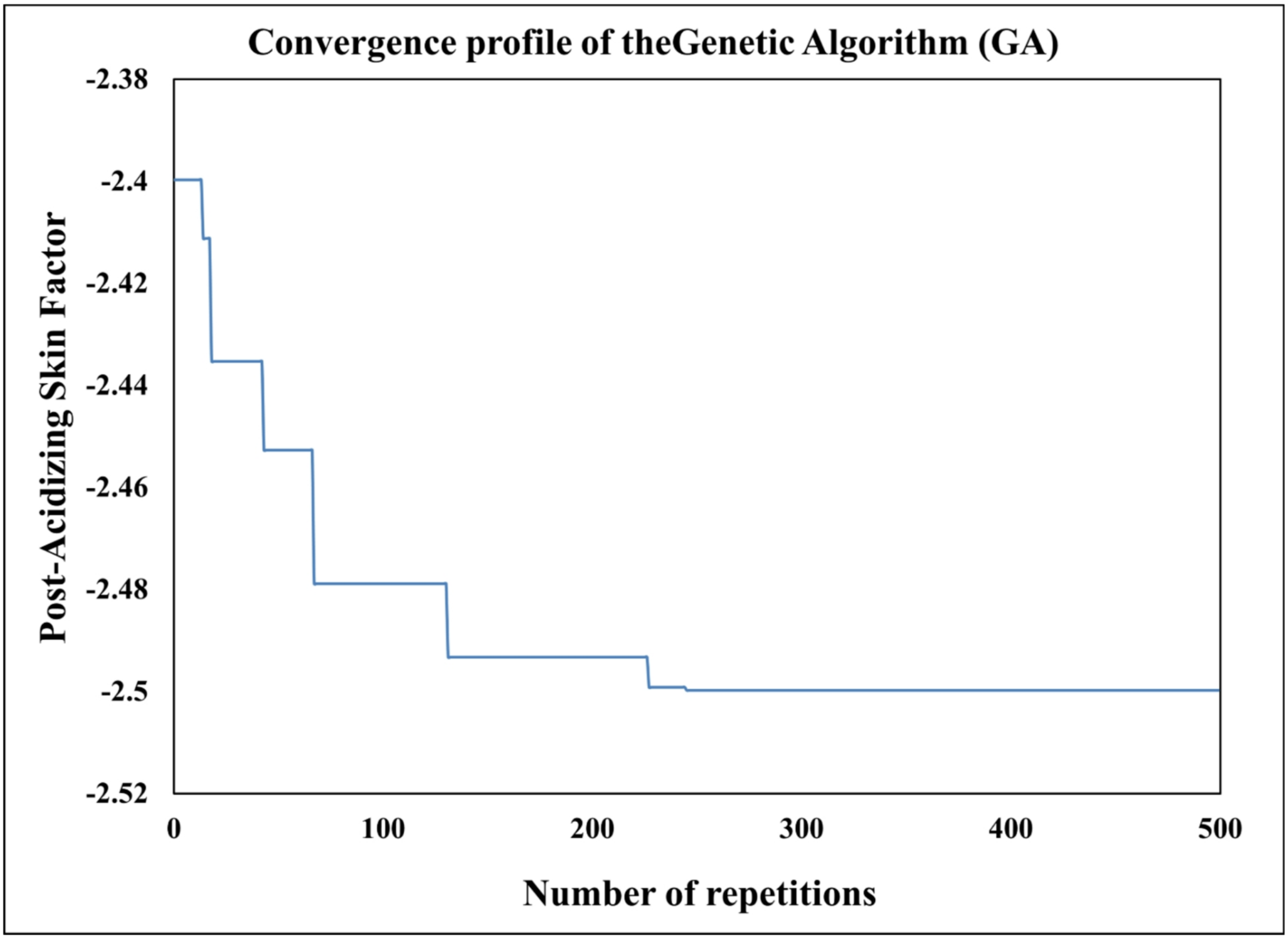

Fig. 14 Convergence profile of the genetic algorithm (GA), illustrating the minimization of the post-acidizing skin factor objective function over 500 iterations

and simpler to implement, but can be susceptible to premature convergence and less robust to noise and local optima.

As shown in Fig. 16 the optimal post-acidizing skin factors determined by GA and PSO are remarkably similar. However, the Genetic Algorithm is selected as the preferred optimization technique due to its superior robustness to noise and its ability to handle complex, non-linear relationships. This choice is further supported by the observed convergence behavior and the demonstrated ability of the GA to effectively explore the solution space and converge to near-optimal solutions.

To assess the performance of the proposed genetic algorithm, a comparison was made with an existing commercial software. Both methods were used to calculate the optimal acid injection rates and volumes for 10 wells.

Figure 17 demonstrates the superior performance of the genetic algorithm. The post-acidizing skin factors obtained from the GA-optimized parameters consistently exhibit lower values compared to those calculated by the commercial software. This indicates that the GA, with its ability to explore a wider solution space and optimize complex, non-linear relationships, can provide more accurate and efficient acid treatment recommendations.

Limitations of predictive and optimization models in Real-World applications

Predictive and optimization models, while powerful tools, have several limitations when applied to real-world conditions:

i. **Data limitations**:

- Data quality: Real-world data often suffers from issues such as missing values, inconsistencies, noise, and biases. These issues can significantly impact model accuracy and reliability by introducing spurious correlations, creating sampling bias, and preventing the model from learning the true underlying physical relationships.

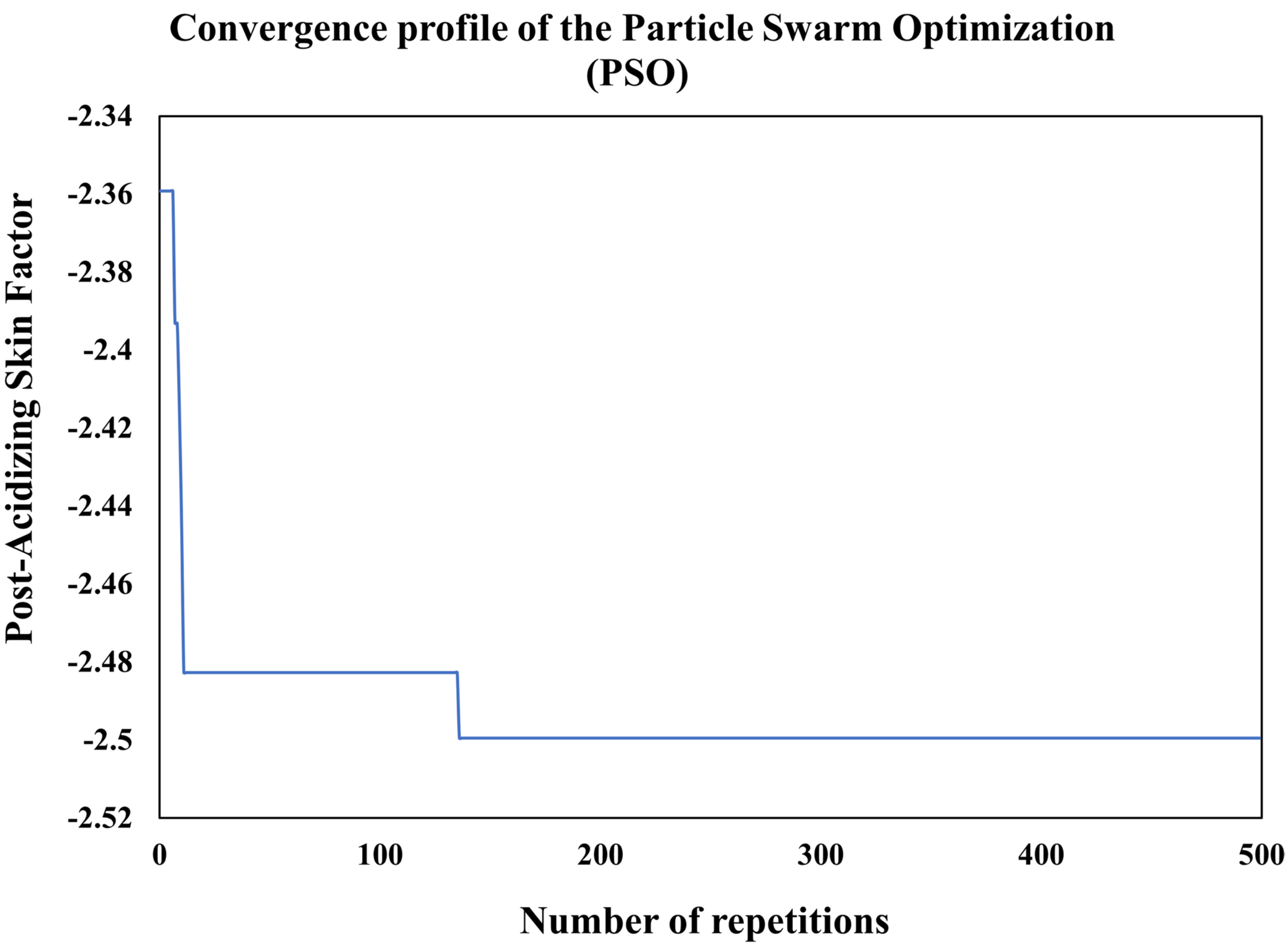

Fig. 15 Convergence profile of the particle swarm optimization (PSO), demonstrating the minimization of the post-acidizing skin factor over 500 iterations

- Data availability:
  Obtaining sufficient, high-quality data for model training and validation can be challenging and time-consuming, especially for complex systems.
- Data representativeness:
  The data used to train the model may not accurately represent the full range of conditions encountered in real-world applications, especially when dealing with complex physical phenomena such as phase transitions in sediments, potentially leading to inaccurate predictions and suboptimal solutions (Gan et al. 2025).
- Dataset size and generalizability:
  Dataset Size and Generalizability: A notable limitation of the presentstudy is the size of the dataset used for model development. Theanalysis was conducted on a dataset of 441 wells, which, whilesubstantial, may be considered modest for training a machine learningmodel intended for broad generalization. Models trained on datasets ofthis size risk overfitting to the specific geological and operational characteristics of the sample, potentially limiting their predictiveaccuracy when applied to wells from different fields. To enhance themodel's robustness and ensure its wider applicability, future researchshould focus on expanding the dataset. Incorporating a larger and morediverse collection of well data would be crucial for developing a moreglobally applicable predictive tool and improving confidence in itsperformance across a wider range of real-world scenarios.

ii. **Model limitations**:

- Model assumptions: Most models rely on simplifying assumptions about the system being modeled. These assumptions may not always hold true in real-world scenarios, leading to discrepancies between model predictions and actual outcomes (Jiang et al. 2024).
- Overfitting: Models can sometimes overfit to the training data, meaning they perform well on the training data but poorly on new, unseen data. This can lead to inaccurate predictions and unreliable optimization results.

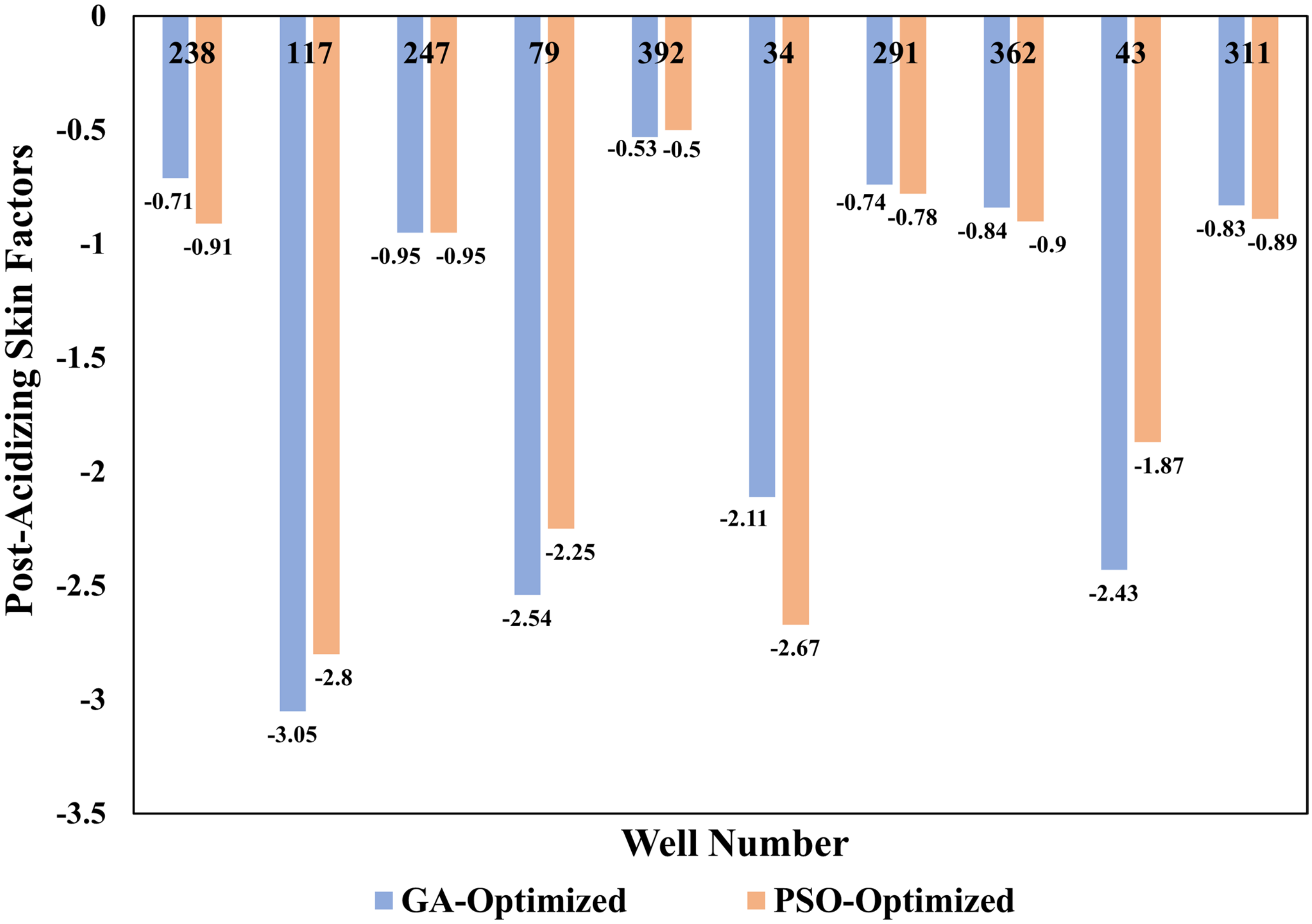

Fig. 16 Comparative performance of the GA- and PSO-optimized models. The bar chart displays the final post-acidizing skin factor for each well, showing that the GA consistently achieved a lower (more negative) skin factor

- Lack of interpretability: Some complex models, such as deep neural networks, can be difficult to interpret. This lack of transparency can hinder understanding of how the model arrives at its predictions and make it difficult to identify and address potential biases or errors.

iii. **Real-world complexity**:

- Unforeseen events: Real-world systems are often subject to unforeseen events and disturbances that are not captured in the model. These events can significantly impact the system's behavior and invalidate model predictions.
- Dynamic environments: Many real-world systems are dynamic and constantly evolving. Models trained on historical data may not be able to accurately predict or optimize performance in changing conditions.
- Human factor: Human behavior and decision-making can significantly influence the performance of any system. Models may not adequately account for these human factors, leading to unexpected outcomes.

iv. **Absence of uncertainty quantification**:

- A key limitation of the current study is that the machine learning models provide point predictions without an associated measure of confidence. To enhance the practical applicability of these predictions, a formal uncertainty quantification (UQ) is necessary. Future work should focus on implementing methods such as bootstrap resampling or Monte Carlo simulations to generate prediction intervals. These techniques would quantify the uncertainty in the model's outputs, providing a range of likely outcomes rather than a single value, thereby offering a more complete risk assessment for operational decision-making.

Conclusions

This study successfully developed and implemented a hybrid machine learning and metaheuristic optimization framework to determine the optimal acid injection parameters for

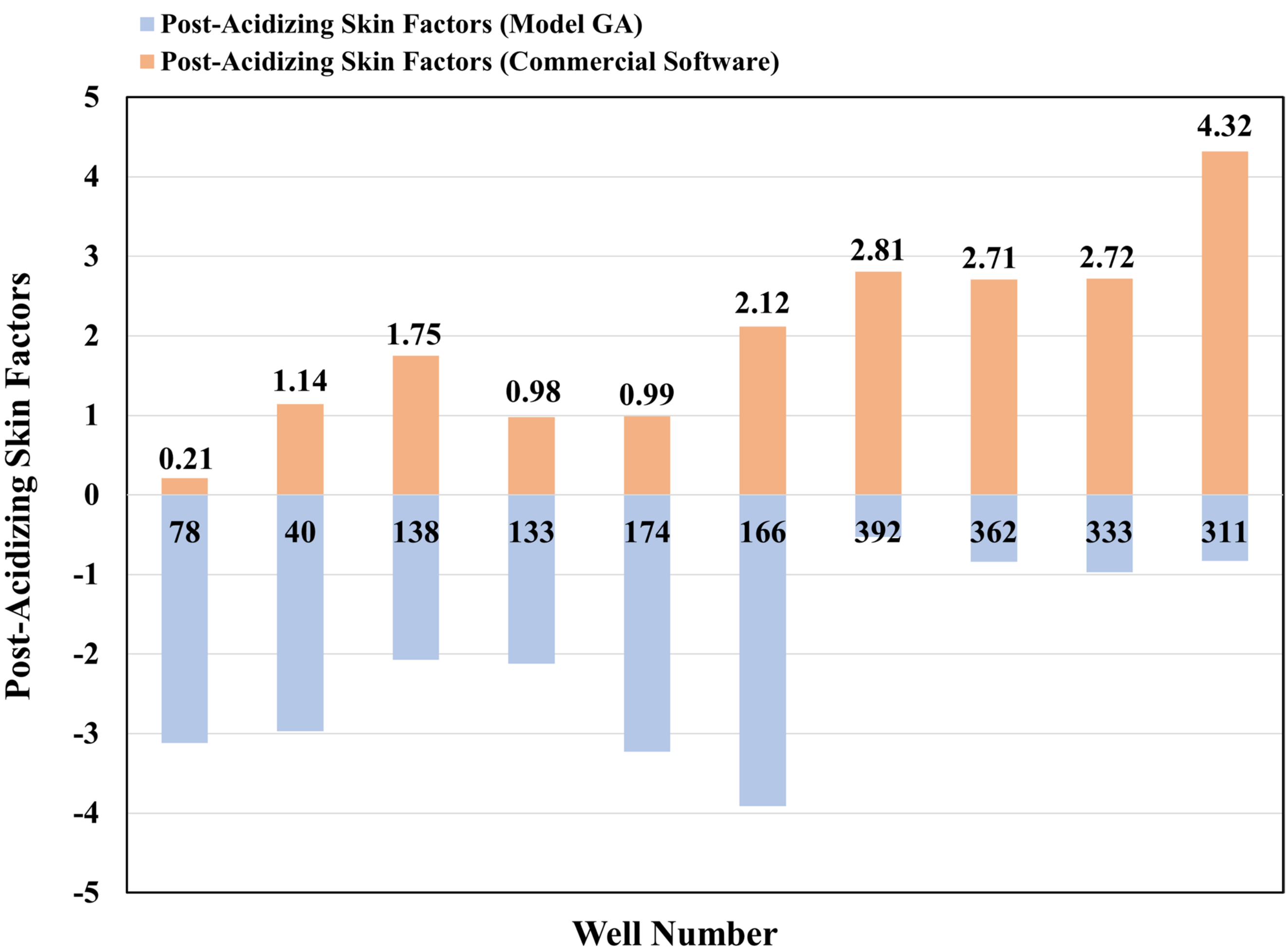

Fig. 17 Comparison of post-acidizing skin factors from treatment parameters optimized by the proposed genetic algorithm (GA) versus those calculated by a commercial software package. The results demonstrate the superior performance of the GA model

minimizing the post-acidizing skin factor. A comparative analysis identified the Extra Trees algorithm as the most robust predictive model for this application. The subsequent optimization phase and overall findings are summarized as follows:

i. **Comparative performance of optimization algorithms**: Both the Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) proved to be highly effective metaheuristic strategies for this optimization problem. The treatment parameters recommended by both algorithms led to significant and broadly comparable reductions in the post-acidizing skin factor across the selected wells. While the GA yielded a nominally better outcome in several cases, the performance differences were generally minor and not consistently in favor of one algorithm. A formal statistical analysis would be required to determine if these small variations are statistically significant. Therefore, based on the available evidence, this study concludes that both GA and PSO are viable and powerful tools for this application, without a clear basis to declare one as definitively superior.
ii. **Superiority over commercial Software baseline**: The optimization framework, particularly when coupled with the Genetic Algorithm, consistently identified injection strategies (acid volume and rate) that yielded a more effective stimulation design than the baseline parameters derived from the commercial software used for comparison. This highlights the potential of data-driven hybrid models to outperform conventional design approaches.
iii. The GA and PSO algorithms were implemented with 500 iterations and an initial population of 200. Additionally, the convergence graph showed that the calculated values exhibit good convergence.

Declarations

Conflict of interest All authors declare that there is no conflict of interest.

Ethical approval All authors confirm that this paper has not been previously published and that the manuscript reflects our own research and analysis in a truthful and complete manner.

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